

BAIO LEARNING SYSTEM

PU-II / NEET BIOLOGY HIGH-YIELD REVISION NOTES

CHAPTER 4: PRINCIPLES OF INHERITANCE AND VARIATION

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1. INTRODUCTION TO GENETICS & HISTORICAL MILESTONES

Genetics is the branch of biological science dealing with **inheritance** (the process by which traits are passed from parents to offspring) and **variation** (the degree by which progeny differs from parents).

Early Understanding of Variation (8000–1000 B.C.): Ancient humans recognized that variations are linked to sexual reproduction. They utilized artificial selection and selective breeding from ancestral wild populations to generate superior domesticated breeds.

Classic NCERT Example: Domestication of Indian milch cattle — **Sahiwal cows in Punjab** developed from ancestral wild cattle.

2. ESSENTIAL GENETIC TERMINOLOGY

- **Gene (Factor):** Functional unit of heredity carrying DNA instructions for a character. Mendel termed them 'factors'.
- **Alleles:** Slightly different contrasting forms of the same gene residing at the same locus on homologous chromosomes (e.g., T and t).
- **Locus:** The specific physical location/coordinate of a gene on a chromosome.
- **Phenotype:** Observable physical or biochemical manifestation of an individual (e.g., Tall vs Dwarf).
- **Genotype:** Genetic/allelic makeup of an organism (e.g., TT , Tt , tt).
- **Homozygous:** Identical alleles at a locus on homologous chromosomes (TT or tt) — true breeding.
- **Heterozygous:** Dissimilar paired alleles at a locus (Tt) — hybrid condition.
- **Dominant Allele:** Expresses in both homozygous (TT) and heterozygous (Tt) states.
- **Recessive Allele:** Expresses exclusively in homozygous state (tt); masked by dominant allele.
- **Hemizygous:** Having only one single allele copy (e.g., X-linked genes in human males, X^hY).

3. MENDEL'S EXPERIMENTS ON GARDEN PEA (*PISUM SATIVUM*)

Gregor Johann Mendel (1822–1884), the *Father of Genetics*, conducted hybridization experiments on *Pisum sativum* for 7 years (1856–1863). He applied statistical analysis and mathematical logic with large sample sizes.

Reasons for Choosing Garden Pea: (i) Distinct contrasting characters, (ii) Short life cycle (annual), (iii) High seed yield (large sample size), (iv) Natural self-pollination with accessible floral structure for artificial emasculation & bagging, (v) Easy cultivation.

Mendel's 7 Pairs of Contrasting Characters (14 True-Breeding Lines):

S.No	Character	Dominant Trait	Recessive Trait
1	Stem Height	Tall (T)	Dwarf (t)
2	Flower Colour	Violet (W / P)	White (w / p)
3	Flower Position	Axial (A)	Terminal (a)
4	Pod Shape	Inflated / Full (I)	Constricted (i)
5	Pod Colour	Green (G)	Yellow (g)
6	Seed Shape	Round (R)	Wrinkled (r)
7	Seed Colour	Yellow (Y)	Green (y)

4. INHERITANCE OF ONE GENE (MONOHYBRID CROSS) & MENDEL'S LAWS

Mendel crossed pure tall (TT) × dwarf (tt). The F_1 was 100% tall (Tt). Selfing F_1 yielded tall and dwarf plants in F_2 without blending.

F_2 Phenotypic Ratio = 3 Tall : 1 Dwarf (3 : 1) | F_2 Genotypic Ratio = 1 TT : 2 Tt : 1 tt (1 : 2 : 1)

Binomial Expansion Derivation: Gamete frequencies from F_1 (Tt) are $\frac{1}{2} T$ and $\frac{1}{2} t$.

F_2 genotype = $(ax + by)^2 = (\frac{1}{2} T + \frac{1}{2} t)^2 = \frac{1}{4} TT + \frac{1}{2} Tt + \frac{1}{4} tt \rightarrow \text{Ratio} = 1 : 2 : 1$

Mendel's Principles Formulated from Monohybrid Cross:

- 1. Law of Dominance (First Law):** (a) Characters are controlled by discrete units called factors occurring in pairs. (b) In a dissimilar pair of factors (Tt), one member dominates (dominant allele) over the other (recessive allele).
- 2. Law of Segregation / Purity of Gametes (Second Law — Universally Valid):** Alleles do not blend. During meiosis/gametogenesis, homologous chromosome pairs and their alleles segregate such that each gamete receives only **one** allele with equal 50% probability.

5. TEST CROSS, BACK CROSS & PUNNETT SQUARE

Punnett Square (Checkerboard): Invented by **Reginald C. Punnett** to calculate the genotypic probability of offspring by plotting male and female gametes on top and side axes.

Types of Crosses:

- **Back Cross:** Crossing F_1 hybrid with *either* dominant or recessive parent.
- **Out Cross:** F_1 (Tt) × Dominant Parent (TT) → 100% Dominant (Tall).
- **Test Cross:** Crossing an individual of dominant phenotype (unknown genotype TT or Tt) with the **homozygous recessive parent** (tt).

Test Cross Outcomes & Significance:

- If unknown is TT : $TT \times tt \rightarrow 100\%$ Tall (Homozygous Dominant).
- If unknown is Tt : $Tt \times tt \rightarrow 50\%$ Tall : 50% Dwarf → **Ratio 1 : 1** (Heterozygous).
- **Monohybrid Test Cross Ratio: 1 : 1**
- **Dihybrid Test Cross Ratio: $RrYy \times rryy \rightarrow 1 : 1 : 1 : 1$**

6. NON-MENDELIAN GENETICS / DEVIATIONS FROM MENDELISM

A. Incomplete Dominance (Partial / Blending-like Expression)

Discovered by **Carl Correns (1903)** in *Mirabilis jalapa* (Four O'clock plant) and *Antirrhinum majus* (Snapdragon / Dog flower). Also seen in Andalusian fowls (Black × White → Blue fowls).

- Heterozygote (Rr) shows an intermediate phenotype (Pink flowers) between Red (RR) and White (rr).

Key NEET Point: In Incomplete Dominance, **Phenotypic Ratio = Genotypic Ratio = 1 : 2 : 1** (1 Red : 2 Pink : 1 White).

B. Co-Dominance & Multiple Allelism

Co-Dominance: Both alleles in a heterozygote express completely and simultaneously without blending (e.g., AB blood group, Roan coat in cattle, $Hb^A Hb^S$ carrier).

Multiple Allelism: Presence of more than 2 alternative alleles of a single gene in a population (e.g., ABO blood system).

ABO Blood Group Genetics (Karl Landsteiner; Bernstein):

- Governed by gene I (Isoagglutinin) with 3 alleles: I^A , I^B , i .
- I^A and I^B synthesize specific sugar polymers (antigens A & B) on RBC membranes; i produces no sugar.
- I^A and I^B are dominant over i , but **co-dominant** with each other ($I^A I^B \rightarrow$ AB Blood).
- Number of Genotypes = 6, Number of Phenotypes = 4 [Formula: $n(n+1)/2 = 3(4)/2 = 6$].

Genotype	Allele P1	Allele P2	RBC Antigen	Plasma Antibody	Blood Group
$I^A I^A / I^A i$	I^A	I^A or i	Antigen A	Anti-B	A
$I^B I^B / I^B i$	I^B	I^B or i	Antigen B	Anti-A	B
$I^A I^B$	I^A	I^B	Antigen A & B	None	AB (Universal Recipient)
ii	i	i	No Antigen	Anti-A & Anti-B	O (Universal Donor)

C. Dominance vs Incomplete Dominance vs Co-Dominance

Feature	Complete Dominance	Incomplete Dominance	Co-Dominance
F₁ Phenotype	Resembles dominant parent	Intermediate phenotype	Resembles both parents
F₂ Phenotypic Ratio	3 : 1	1 : 2 : 1	1 : 2 : 1
F₂ Genotypic Ratio	1 : 2 : 1	1 : 2 : 1	1 : 2 : 1
Example	Pea height ($TT \times tt$)	Snapdragon flower color (Rr)	AB blood group; Roan cattle (RW)

D. Pleiotropy vs Polygenic Inheritance

Pleiotropy (1 Gene → Multiple Traits):

A single gene influences multiple phenotypic traits by regulating a common metabolic pathway.

- **Phenylketonuria (PKU):** Mutation in *PAH* gene (chromosome 12) → mental retardation, skin hypopigmentation, hair loss.
- **Starch Synthesis in Pea:** Gene *B* controls starch synthesis and seed shape. *BB* (round, large grains), *bb* (wrinkled, small grains), *Bb* (round seeds, intermediate grain size).

Polygenic Inheritance (Many Genes → 1 Trait):

Quantitative trait controlled by multiple additive genes, influenced by environmental factors.

- **Human Skin Colour (Galton / Davenport):** Controlled by 3 genes (*A, B, C*). *AABBCC* (Darkest) × *aabbcc* (Lightest) → F₂ bell curve: **1:6:15:20:15:6:1** (64 combinations).
- **Kernel Colour in Wheat:** First proven by Nilsson-Ehle (Ratio 1:4:6:4:1).

7. DIHYBRID CROSS & CHROMOSOMAL THEORY OF INHERITANCE

Mendel crossed round yellow ($RRYY$) × wrinkled green ($rryy$). F₁ was 100% round yellow ($RrYy$). Selfing F₁ gave 16 combinations.

F₂ Phenotypic Ratio = 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green (9 : 3 : 3 : 1)

F₂ Genotypic Ratio = 1:2:1:2:4:2:1:2:1 (9 Distinct Genotypes)

Law of Independent Assortment (Third Law): When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the segregation of the other pair during gamete formation.

Rediscovery of Mendel's Work (1900): Independent rediscovery by **Hugo de Vries** (Holland), **Carl Correns** (Germany), and **Erich von Tschermak** (Austria).

Chromosomal Theory of Inheritance (Sutton & Boveri, 1902): Parallelism between genes and chromosomes — both occur in pairs, segregate at Anaphase I of meiosis, and independent pairs assort independently.

8. LINKAGE AND RECOMBINATION (T.H. MORGAN)

Thomas Hunt Morgan (the *Father of Experimental Genetics*) proved chromosomal theory using *Drosophila melanogaster* (fruit fly).

Advantages of *Drosophila*: Reared on simple synthetic medium, 2-week life cycle, single mating yields hundreds of flies, clear sexual dimorphism, distinct morphological markers visible under low power microscope.

- **Linkage:** Physical association of genes on the same chromosome. Linked genes do not assort independently and deviate from the 9:3:3:1 ratio.
- **Recombination:** Non-parental gene combinations arising from crossing over during Pachytene of Meiosis I.
- **Morgan's Crosses:**
 - *Cross A (Body colour & Eye colour - y and w):* Tightly linked → **98.7% Parental : 1.3% Recombinant.**
 - *Cross B (Eye colour & Wing size - w and m):* Loosely linked → **62.8% Parental : 37.2% Recombinant.**
- **Alfred Sturtevant:** Used recombination frequencies to create the first **Genetic Maps** (1% recombination = 1 cM = 1 map unit).

9. SEX DETERMINATION MECHANISMS

Hermann Henking (1891) identified the '**X-body**' in insect spermatogenesis, later identified as the X chromosome.

Mechanism	Male	Female	Examples
XX - XY (Male Heterogamety)	XY (50% X, 50% Y)	XX (100% X)	Humans, <i>Drosophila</i> , mammals
XX - XO (Male Heterogamety)	XO (50% X, 50% null)	XX (100% X)	Grasshoppers, Insects
ZZ - ZW (Female Heterogamety)	ZZ (Homogametic)	ZW (50% Z, 50% W)	Birds, Fowls, Reptiles, Fishes
ZZ - ZO (Female Heterogamety)	ZZ (Homogametic)	ZO (Heterogametic)	Butterflies, Moths

Haplodiploidy in Honey Bees (*Apis mellifera*):

- **Females (Queen/Worker):** Diploid (**2n = 32**) formed by fertilization of egg (n=16) and sperm (n=16).
- **Males (Drones):** Haploid (**n = 16**) formed unfertilized via **Parthenogenesis**.
- Drones produce sperms by **mitosis**. Therefore, male honeybees **have no father and no sons, but have a grandfather and grandsons!**

10. MUTATION & PEDIGREE ANALYSIS

Mutation: Sudden inheritable change in DNA sequence.

- **Point Mutation:** Single base change (e.g., Sickle-cell anemia: GAG → GUG).
- **Frameshift Mutation:** Deletions or insertions of bases shifting reading frames.
- **Mutagens:** Physical (UV rays, X-rays), Chemical (Nitrous acid, EMS), Biological (Viruses).

Pedigree Analysis Key Rules:

- **Autosomal Dominant:** Appears vertically in every generation; unaffected parents do not transmit trait (e.g., Myotonic dystrophy).
- **Autosomal Recessive:** Trait skips generations; carriers are unaffected; consanguinity increases frequency (e.g., Sickle cell anemia, PKU).
- **X-Linked Recessive:** Criss-cross inheritance (Grandfather → Carrier daughter → Grandson); far more common in males (e.g., Haemophilia, Colour blindness).

11. HUMAN GENETIC DISORDERS

A. Mendelian Disorders (Gene Mutations)

Disorder	Inheritance	Chromosome	Pathology & Clinical Features
Sickle Cell Anaemia	Autosomal Recessive ($Hb^A Hb^S$)	Chr 11 (<i>HBB</i> gene)	Glutamic acid → Valine substitution at 6 th position of β -globin chain. Sick RBCs cause hemolysis and microvascular occlusion. Heterozygotes show malaria resistance.
Thalassemia	Autosomal Recessive (Quantitative defect)	α : Chr 16 (<i>HBA1,2</i>) β : Chr 11 (<i>HBB</i>)	Reduced synthesis of α or β globin chains → severe hemolytic anemia, microcytosis, and splenomegaly.
Phenylketonuria (PKU)	Autosomal Recessive (Inborn error)	Chr 12	Deficiency of <i>phenylalanine hydroxylase</i> → accumulation of phenylpyruvate → mental retardation, hypopigmentation.
Haemophilia (Royal Disease)	X-Linked Recessive (Criss-cross)	X Chromosome	Deficiency of Factor VIII (Haemophilia A) or Factor IX (Haemophilia B). Failure of blood clotting → fatal hemorrhage from minor cuts.
Colour Blindness	X-Linked Recessive (Daltonism)	X Chromosome	Defect in red or green cone photopigments → red-green color blindness. Affects ~8% males, ~0.4% females.
Myotonic Dystrophy	Autosomal Dominant	Chr 19 (<i>DMPK</i>)	Trinucleotide (CTG) repeat expansion → muscle wasting, myotonia, cataracts.

B. Chromosomal Disorders (Aneuploidies)

Disorder	Karyotype & Genetic Basis	Clinical Features
Down's Syndrome	47, XX/XY,+21 (Trisomy 21; Langdon Down 1866)	Short stature, rounded flat face, furrowed tongue, partially open mouth, Simian palm crease, congenital heart disease, mental retardation.
Klinefelter's Syndrome	47, XXY (Allosomal Trisomy in Males)	Tall eunuchoid male, sterile, azoospermia, gynecomastia (breast development), sparse body hair.
Turner's Syndrome	45, XO (45, X) (Allosomal Monosomy in Females)	Sterile female, rudimentary/streak ovaries, short stature, webbed neck, shield-shaped chest, lack of secondary sexual characters.
Cri-du-Chat Syndrome	46, 5p- (Partial deletion of short arm of Chr 5)	Cat-like high-pitched cry due to laryngeal dysgenesis, microcephaly, mental retardation.

12. HIGH-YIELD PRACTICE PROBLEMS (SOLVED)

Problem 1: A child has blood group 'O'. If the father is blood group 'A' and mother is blood group 'B', find the genotypes of parents and offspring ratios.

Solution: Child is ii (O). Hence, both parents must carry allele i .

Father's genotype = $I^A i$; Mother's genotype = $I^B i$.

Possible offspring: $I^A I^B$ (AB) : $I^A i$ (A) : $I^B i$ (B) : ii (O) in 1 : 1 : 1 : 1 ratio (25% probability each).

Problem 2: A carrier haemophilic woman ($X^H X^h$) marries a normal man ($X^H Y$). What percentage of male children will have haemophilia?

Solution: Male gametes: X^H , Y; Female gametes: X^H , X^h . Sons receive Y from father and either X^H or X^h from mother → 50% $X^H Y$ (Normal) : 50% $X^h Y$ (Haemophilic).

Problem 3: In a dihybrid test cross, if recombinant frequency is 18%, what is the distance between the two linked genes?

Solution: 1% Recombination Frequency = 1 CentiMorgan (cM) = 1 Map Unit (m.u.). Therefore, the map distance is **18 cM** (or 18 map units).

BAIO LEARNING SYSTEM — RAPID RECAP

- **Mendelism:** Monohybrid (Phenotypic 3:1, Genotypic 1:2:1); Dihybrid (9:3:3:1).
- **Exceptions:** Incomplete Dominance (1:2:1), Co-dominance (ABO blood group), Linkage (Morgan in *Drosophila*).
- **Sex Determination:** Humans (XX-XY), Birds (ZZ-ZW), Insects (XX-XO), Honeybee (Haplodiploidy 32/16).
- **Key Disorders:** Sickle Cell (Glu → Val at pos 6 of β -chain), Down's (Trisomy 21), Klinefelter's (XXY), Turner's (XO).